



**ALLIANCE
UNIVERSITY**
Private University established in Karnataka State by Act No. 34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 8

Title: MARKETING CAMPAIGN PERFORMANCE DASHBOARD

Submitted By:

Name: Vallepu Balaji

Reg. No.: 2511022250077

Date: 30/08/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

EXPERIMENT NO : 8

MARKETING CAMPAIGN PERFORMANCE DASHBOARD USING POWER BI

Problem Statement

You are a Marketing Analyst in a digital marketing agency. Multiple marketing campaigns have been executed across different channels, campaign types, customer segments, and locations. Management wants to evaluate campaign performance based on important marketing indicators such as impressions, clicks, conversion rate, acquisition cost, engagement score, and Return on Investment (ROI).

The objective of this project is to develop an interactive **Marketing Campaign Performance Dashboard using Microsoft Power BI**. The dashboard provides a centralized view of campaign performance and enables management to compare different campaigns, channels, customer segments, and locations.

The dashboard is designed with two interactive pages. The first page provides an executive overview of overall campaign performance, while the second page provides detailed campaign, conversion, cost, engagement, and ROI analysis.

Methodology / Steps Followed

Step 1: Data Collection

The marketing campaign dataset was collected in CSV format and imported into Power BI for analysis.

The dataset contains campaign-level information such as:

Field	Description
Campaign_ID	Unique campaign identification
Company	Company associated with the campaign
Campaign_Type	Type of marketing campaign
Target_Audience	Target customer group
Duration	Campaign duration
Channel_Used	Marketing channel used
Conversion_Rate	Percentage of users converted
Acquisition_Cost	Cost incurred to acquire customers

Field	Description
ROI	Return on Investment
Location	Campaign location
Language	Language targeted
Clicks	Number of clicks generated
Impressions	Number of times advertisements were displayed
Engagement_Score	Customer engagement measurement
Customer_Segment	Customer classification
Date	Campaign execution date

The dataset contains **200,000 campaign records and 16 columns**.

Step 2: Data Import

The following steps were followed to import the dataset:

1. Opened **Microsoft Power BI Desktop**.
2. Selected **Home** → **Get Data**.
3. Selected **Text/CSV**.
4. Selected the marketing campaign CSV file.
5. Clicked **Transform Data** to open Power Query.
6. Verified the imported columns and data types.
7. Loaded the cleaned data into Power BI.

The dataset was imported into Power BI as the main **Marketing Campaign** table.

Step 3: Data Cleaning and Transformation

Power Query was used to clean and transform the raw marketing dataset before developing the dashboard.

The following operations were performed:

- Checked for duplicate records.
- Checked missing or invalid values.

- Renamed columns where required.
- Verified data types.
- Converted the Date column into the Date data type.
- Converted Acquisition Cost into a numerical value.
- Removed currency symbols and commas from acquisition cost where necessary.
- Verified Clicks and Impressions as numerical fields.
- Verified Conversion Rate and ROI as numerical fields.
- Checked campaign and channel names for consistency.

The Acquisition Cost field was transformed from values such as \$16,174.00 into a numerical value suitable for calculations.

Step 4: Data Modeling

A Date table was created in Power BI to support time-based analysis.

The Date table included:

- Date
- Year
- Month
- Month Number
- Month Year
- Quarter

A relationship was created between the Date table and the Marketing Campaign table.

The model allows campaign performance to be analyzed by:

- Year
- Month
- Campaign Type
- Channel
- Location
- Customer Segment
- Target Audience

Step 5: DAX Measures

DAX measures were created to calculate important marketing KPIs.

Total Campaigns

Total Campaigns =

DISTINCTCOUNT('Marketing Campaign'[Campaign_ID])

Total Impressions

Total Impressions =

SUM('Marketing Campaign'[Impressions])

Total Clicks

Total Clicks =

SUM('Marketing Campaign'[Clicks])

Estimated Conversions

Since the dataset provides Clicks and Conversion Rate rather than a direct conversion-count field, estimated conversions were calculated as:

Estimated Conversions =

SUMX(
 'Marketing Campaign',
 'Marketing Campaign'[Clicks] *
 'Marketing Campaign'[Conversion_Rate]
)

Average ROI

Average ROI =

AVERAGE('Marketing Campaign'[ROI])

Click Through Rate

CTR =

DIVIDE(
 [Total Clicks],
 [Total Impressions],
 0

)

Cost Per Click

Cost Per Click =

DIVIDE(

[Total Acquisition Cost],

[Total Clicks],

0

)

These measures were used throughout the dashboard to provide dynamic calculations based on the selected filters.

Dashboard Development

The final dashboard was developed using a modern blue, purple, green, orange, and cyan color theme.

Color Theme

Dashboard Element Color Code

Header	#172554
Primary Blue	#2563EB
Impressions	#0891B2
Clicks	#F97316
Conversions	#7C3AED
ROI	#16A34A
Background	#F8FAFC
Cards	FFFFFF
Text	#1E293B
Border	#E2E8F0

The use of consistent colors makes the dashboard visually attractive and helps users quickly understand different KPIs.

PAGE 1: EXECUTIVE CAMPAIGN OVERVIEW

The first dashboard page provides a high-level overview of marketing campaign performance.

KPI Cards

Five KPI cards were created:

1. **Total Campaigns**
2. **Total Impressions**
3. **Total Clicks**
4. **Estimated Conversions**
5. **Average ROI**

Each KPI uses a different accent color to improve visual identification.

- Campaigns – Blue
- Impressions – Cyan
- Clicks – Orange
- Conversions – Purple
- ROI – Green

Monthly Campaign Performance

A line chart was created to analyze campaign performance over time.

The chart displays:

- Impressions
- Clicks
- Estimated Conversions

The visual allows management to identify monthly increases or decreases in campaign activity.

ROI by Campaign Type

A column chart was created to compare the average ROI of different campaign types.

This helps management identify campaign types that provide stronger returns.

ROI by Marketing Channel

A bar chart was used to compare ROI across marketing channels such as Google Ads, Facebook, Instagram, YouTube, Email, and other available channels.

Campaign Distribution by Target Audience

A donut chart was created to display campaign distribution across different target audiences.

Conversion Rate by Customer Segment

A column chart was created to compare conversion rates between different customer segments.

Interactive Filters

The first page contains interactive slicers for:

- Year
- Campaign Type
- Channel
- Location
- Customer Segment

Selecting any slicer automatically updates the KPI cards and charts.

PAGE 2: CAMPAIGN & ROI ANALYSIS

The second dashboard page provides detailed analysis of campaign efficiency, investment, engagement, and conversion.

KPI Cards

Four KPI cards were created:

1. **CTR**
2. **Conversion Rate**
3. **Cost Per Click**
4. **Cost Per Conversion**

These KPIs provide detailed information about campaign efficiency.

ROI vs Acquisition Cost

A scatter chart was created to compare:

- Acquisition Cost
- ROI
- Clicks
- Marketing Channel

This visual helps identify campaigns that generate high ROI with relatively lower acquisition costs.

Channel Effectiveness

A combination chart was used to compare marketing channels based on:

- Average ROI
- Conversion Rate

This helps management determine which channels provide the best overall performance.

Conversion Rate by Campaign Type

A column chart was created to compare conversion rates among different campaign types.

Geographic Campaign Performance

A map visual was included to analyze campaign performance across different locations.

The map can display:

- Impressions
- ROI
- Conversion Rate
- Clicks

This allows management to identify stronger and weaker geographic markets.

Engagement Score vs Conversion Rate

A scatter chart was created to compare Engagement Score with Conversion Rate.

This helps determine whether campaigns with stronger customer engagement also generate better conversion results.

Detailed Campaign Performance Table

A detailed table was created containing:

- Campaign ID
- Company
- Campaign Type
- Channel
- Target Audience
- Clicks
- Impressions

- Conversion Rate
- ROI
- Acquisition Cost
- Engagement Score

Conditional formatting was applied to ROI and Conversion Rate to highlight high- and low-performing campaigns.

Interactive Dashboard Features

The dashboard was designed to be interactive rather than static.

The following interactive features were implemented:

Slicers

Users can filter the complete dashboard using:

- Year
- Campaign Type
- Channel
- Location
- Customer Segment

Page Navigation

A page navigator was added to move between:

Executive Overview

and

Campaign & ROI Analysis

Reset Filters

A Reset Filters button was created using a Power BI bookmark.

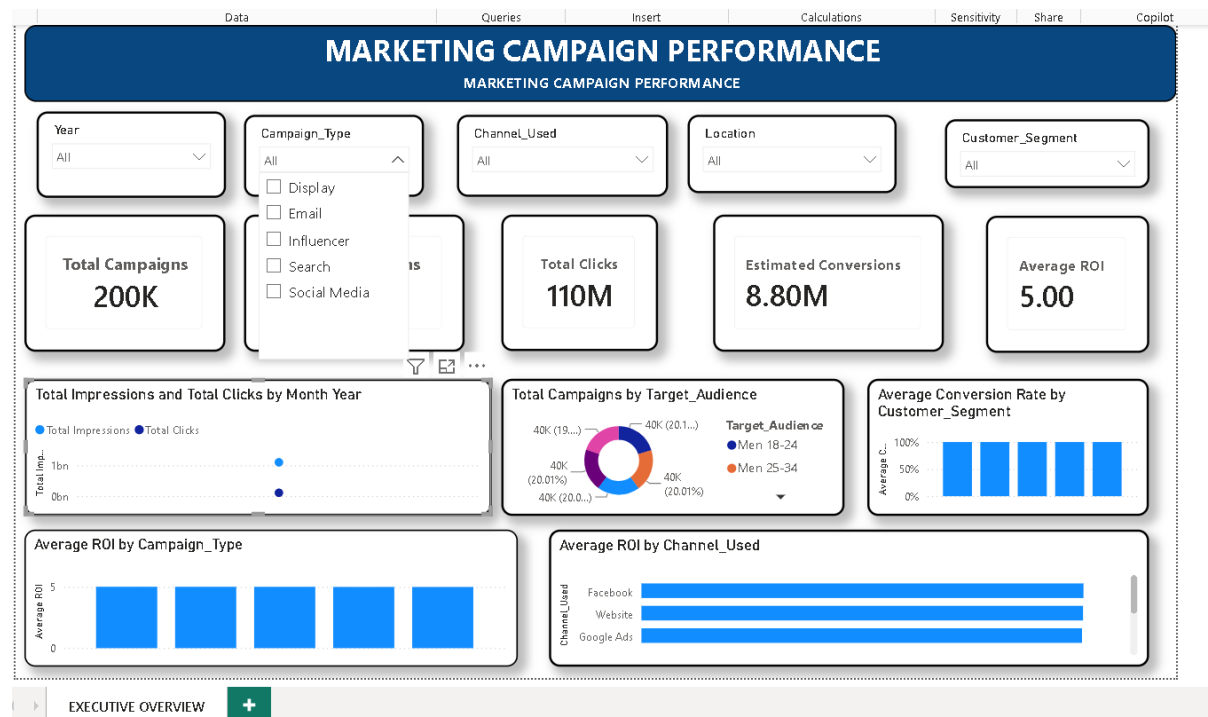
Cross Filtering

Selecting a campaign type, channel, location, or customer segment automatically filters the other dashboard visuals.

Tooltips

Additional campaign information can be displayed when the user hovers over charts and data points.

Output



Conclusion

This project successfully developed an interactive **Marketing Campaign Performance Dashboard using Microsoft Power BI**. The raw CSV marketing dataset was imported, cleaned, transformed, and modeled for business analysis.

Power Query was used for data cleaning and transformation, while DAX was used to create important marketing KPIs such as Total Campaigns, Total Impressions, Total Clicks, Estimated Conversions, CTR, Cost Per Click, Cost Per Conversion, and Average ROI.

The dashboard was divided into two pages. The first page provides an executive overview of campaign performance, while the second page provides detailed ROI, cost, engagement, conversion, channel, and geographic analysis.

The use of interactive slicers, charts, KPI cards, page navigation, conditional formatting, and tooltips makes the dashboard user-friendly and suitable for management decision-making.

Overall, the project demonstrates how Power BI can transform raw marketing data into meaningful visual insights and support data-driven marketing decisions.

Learning Outcomes

Through this project, I learned:

- How to import CSV datasets into Power BI.

- How to use Power Query for data cleaning and transformation.
- How to identify and handle incorrect data types.
- How to transform currency and numerical fields.
- How to create a Date table in Power BI.
- How to create relationships between tables.
- How to create DAX measures and calculated metrics.
- How to calculate marketing KPIs such as CTR, ROI, and Cost Per Click.
- How to create interactive KPI cards.
- How to develop line charts, bar charts, column charts, donut charts, maps, scatter charts, and tables.
- How to apply professional colors and formatting.
- How to create slicers and interactive filters.
- How to implement page navigation.
- How to use bookmarks for Reset Filters.
- How to apply conditional formatting.
- How to analyze campaign performance using Power BI.
- How to convert raw marketing data into meaningful business insights.
- How to design a professional two-page business intelligence dashboard.

Github Link: https://github.com/Balajivallepu/Data_Analytics-_for_Bussiness/tree/main/EXPERIMENT8